

`.broker.zip` / `.status.zip` — wire schema reference

Source of truth: `ai-gateway/protocol/src/lib.rs` (crate `protocol`, linked by both `ai-protect` and `ai-gateway` so a renamed field is a compile error on both ends, not a silent wire mismatch). This doc is a reading aid, not a replacement — re-check the source before relying on a field name or default.

Current wire version: `BROKER_SCHEMA_VERSION = "1"`. Bumped only on a breaking change; an additive field an old reader ignores does not bump it.

Transport model

No persistent socket. Each real user's gateway home (`~/zsai-gateway` by default, `$ZSAI_GATEWAY_HOME`) holds two gzipped-JSON files, each side watching the one it reads:

File	Writer	Reader	Contract
<code>.broker.zip</code>	ai-protect	ai-gateway (service mode)	<code>BrokerFile</code> — full snapshot every write
<code>.status.zip</code>	ai-gateway (service mode)	ai-protect	<code>StatusFile</code> — full snapshot every write

Full snapshot, not a change feed, on both sides. Neither file is ever a diff or an append. `.broker.zip`'s `agents[]` is ai-protect's complete current discovery list every time; `.status.zip`'s `agents[]` carries every agent's complete current wiring every time, including one already reported healthy on the previous write. This is why the merge on the ai-protect side (`ConfigDelta`, below) has to be idempotent — the same delta arrives on every write for as long as that agent is serviceable, not once.

Absent means empty, never missing. Every string field is present on the wire and carries `""` when unknown — a reader never has to distinguish "omitted" from "no value" — and every field carries `#[serde(default)]` regardless, so a writer built before a field existed still decodes.

`BrokerFile` — ai-protect → ai-gateway

```
BrokerFile {
  schema_ver: String,           // "1"; "" from a pre-field writer is treated as a mismatch
  tenant: Tenant,
  credentials: UserCredentials,
  platform: Platform,
  device: Device,
  provenance: Provenance,
  agents: Vec<AgentEntry>,      // full current discovery list, never an append
}
```

- `tenant.cloud` — cloud/endpoint selector; `""` ⇒ ai-gateway's compiled-in default. **A changed value rebuilds the whole `AgentManager`** (see the flow doc) — it isn't just consulted, it forces the endpoints, OIDC, and credential store to be re-derived.
- `tenant.oidc` — `Oidc { issuer, client_id, audience, scope, authorize_path, token_path, discovery_path, tenant_id_claim }`. `""` in any field means "use the SDK default"; a blank `issuer` specifically means ai-gateway cannot run interactive OIDC at all if it needs to.
- `credentials` — `UserCredentials { user_jwt, refresh_token }`. Both blank ⇒ "ai-protect has nothing to give"; ai-gateway falls back to its own cached refresh token, then interactive login. `UserCredentials::available()` is `!user_jwt.is_empty() || !refresh_token.is_empty()`. ⚠ **Currently always blank in practice** — `identity::zax_user_credential::fetch` on the ai-protect side is a stub (no real ZAX-user-JWT source wired yet).
- `platform` / `device` / `provenance` — host facts gathered by ai-protect so ai-gateway does not re-probe; field-for-field the shape the SDK's own enrolment payload expects. `provenance.root_proof` / `root_proof_type` ("none" today) and `provenance.attestation.tpm_present` are wired but not yet populated with anything meaningful.
- `agents[]` — one `AgentEntry` per agent ai-protect's discovery inventory maps to a known broker adapter id (`client_label_to_broker_id` in `src/subscribers/ai_broker_launch.rs`). Written **without diffing**: ai-gateway is the one that dedups/filters against its own `agents::ENABLED_AGENTS` list. `AgentEntry` flattens a `Process` (`name`, `format`, `entry_point`, `language`, `filename`, `runtime`, `cmdline`, `software`, `pid`) plus `description` and an optional `parent: Process` (the launching process, `None` when installed-but-not-running). **Today ai-protect only ever populates `process.name`** (the adapter id string, e.g. "codex") — every other `Process` / `Software` / `CodeSigning` field is the type's default. The full shape exists on the wire (code-signing state, binary hash, publisher, cert chain) for when that richer discovery data is wired through; it is not a schema gap, it is an unfinished populate.

StatusFile — ai-gateway → ai-protect

```
StatusFile {
  message: String,           // human-readable, for logs
  credential: CredentialStatus,
  agents: Vec<AgentStatus>,  // full current wiring, in .broker.zip's agent order
}
```

- `credential.message` — a `CredentialState: Good | Expiring | Expired | Other(String)` (`Other("")` is the default, "nothing reported yet"). `CredentialStatus::needs_refresh()` is true for `Expiring` / `Expired` only. ai-protect edge-triggers on the **false→true transition** of `needs_refresh()` to refetch a credential and rewrite `.broker.zip` — it does not refetch on every snapshot while the state stays `Expiring`, and an unrecognised `Other` value is deliberately never

treated as a refresh request (acting on a value neither side can interpret would turn every unknown string into a credential fetch).

AgentStatus — one row per agent

```
AgentStatus {
  name: String,                // echoed VERBATIM from the .broker.zip entry
  status: AgentRegistrationStatus, // NotSupported | NotApproved | NotAbleToRegister | Success
  aid_key: String,             // "" until registered
  agent_id: String,            // "" until registered
  config: Vec<ConfigDelta>,
  process_env: Vec<ManagedEnvVar>,
  process_env_commands: Vec<String>,
}
```

- **name** is echoed, not normalised. `.broker.zip` may name an agent by adapter id (`"codex"`) or display name (`"Codex CLI"`); `AgentStatus.name` returns whichever form it was sent, because ai-protect correlates rows by that string.
- **status** — only `Success` counts (`is_success()`). The two in-progress values ai-protect's own code treats specially are carried as `Other("enrolling")` / `Other("refreshing")` (the constants `agent_manager::ENROLLING` / `REFRESHING`) — distinguishing a first enrolment from a lapsed one that's re-enrolling.
- **config** — empty until that agent is both enrolled AND currently signable. This is the gate that matters most: a `ConfigDelta` is never emitted for an agent that can't yet be served, specifically so ai-protect never points an agent at a listener with nothing behind it. See `AgentManager::state_of` — `enrolling` or `!is_serviceable()` $\Rightarrow$ empty status with no config; a lapsed triple JWT (`!store.agent_can_sign`) also reports empty, as `Other("refreshing")`.
- **process_env** / **process_env_commands** — the process-env delivery leg (Amp, Cursor's `cursor-agent`, Antigravity's `agy`, Aider's Anthropic-family leg, ...): named OS env vars for an agent with no config-file knob at all. `process_env_commands` names the exact `$PATH` command(s) the vars are scoped to (e.g. `["cursor-agent"]`) — required because two agents can want different values for the same var name (`HTTPS_PROXY`), which a shell-wide `export` can't represent; ai-protect's `src/ai_broker_env/` wraps each named command in its own shell function (Unix) or PowerShell profile function rather than exporting globally. Empty iff `process_env` is empty. **ai-gateway never touches the OS environment itself** — this field only describes intent, same division of labor as `config`.

ConfigDelta — one file, one merge-able change

```
ConfigDelta {
  file: String,    // home-relative, e.g. ".codex/config.toml"
```

```
setting: Value,    // a JSON object merges key-by-key; anything else overwrites
}
```

ai-gateway computes this; **ai-protect is the only writer of the real file**, by design — ai-gateway never touches an agent's own config. Two gates on the ai-protect side that are easy to miss when reasoning about this from the schema alone:

1. **The allowlist** (`gateway_util::config_merge::ALLOWED_CONFIG_PATHS`, 32 entries as of this doc) — `file` must be an exact match or the delta is refused on arrival with "not an allowlisted agent-config path", *even if* the adapter that produced it is otherwise correct. Every new adapter or new settings file needs an entry here — three agents this session (Kiro, Antigravity, Aider) had a *correct* delta silently refused because this list was never updated when their adapter shipped.
2. **The format dispatch** (`gateway_util::config_merge::format_for`, by file extension) — `.toml` merges via `toml_edit` (preserves comments on untouched keys), `.yaml / .yml` merges via `serde_yaml` (comments on untouched keys are **lost** — no AST-preserving edit exists for YAML), anything else merges as JSON via a JSONC-tolerant parser (comments stripped on re-serialize). All three refuse rather than replace on a parse failure — a malformed existing file is left alone, not overwritten with an empty object.

ManagedEnvVar / EnvVarKind

```
ManagedEnvVar { name: String, value: String, kind: EnvVarKind }
EnvVarKind = LoopbackUrl | CaPath | Fixed
```

`kind` is how ai-protect's "never clobber a real pre-existing value" guard decides ownership **without re-deriving policy ai-gateway already computed**:

- `LoopbackUrl` — `value` is expected to be `http(s)://127.0.0.1:*` or `localhost:*`. Ours iff whatever is currently set, if anything, is *also* loopback-shaped.
- `CaPath` — `value` is a filesystem path to the local CA bundle. Ours iff the current value also names it (path suffix match on `certs/ca-cert.pem`).
- `Fixed` — `value` is a literal with no per-install variation (e.g. Cursor's `NODE_USE_ENV_PROXY=1`). Ours iff the current value, if any, already equals it exactly.

What's on the wire but not yet populated

Worth flagging explicitly for anyone reviewing the schema expecting more than ai-protect currently sends:

- `AgentEntry.process.software` (name/version/hash/bundle id/publisher) and `.code_signing` (the full X.509 chain-and-trust shape) — fully defined, always default today. `discovered_broker_agents()` only ever sets `process.name`.

- `AgentEntry.parent` — always `None` ; nothing populates "what launched this agent" onto the wire yet, though ai-protect's own hook-side process lineage walk (used elsewhere for policy) could feed it.
- `Provenance.root_proof` / `attestation.tpm_present` — always blank/false.
- `BrokerFile.credentials` — always blank in practice (see above).

None of these are schema gaps — the fields exist specifically so populating them later is additive, not a wire break.